

Hotel and Restaurant Management System:

Detailed Requirement Analysis

Overview:

The **Hotel and Restaurant Management System** is designed to handle hotel reservations, dining services, inventory management, billing, customer feedback, and event organization. The system aims to streamline hotel operations while ensuring a smooth customer experience.

Entities and Attributes:

1. Customer:

- **Attributes:** customer_id, name, contact_details, email, loyalty_status, address
- **Description:** Manages information about customers who stay at the hotel or dine at the restaurant.

2. Room:

- **Attributes:** room_id, room_number, type, price_per_night, status, amenities
- **Description:** Tracks the details and status of hotel rooms (e.g., available, occupied, under maintenance).

3. Reservation:

- **Attributes:** reservation_id, customer_id (FK), room_id (FK), check_in_date, check_out_date, total_price, status
- **Description:** Manages room reservations and associated details like duration, room type, and customer.

4. Staff:

- **Attributes:** staff_id, name, role, contact_details, hire_date, salary
- **Description:** Tracks the hotel staff members, their roles (e.g., receptionist, waiter), and other employment details.

5. FoodItem:

- **Attributes:** food_item_id, name, price, type (veg/non-veg), availability_status
- **Description:** Manages the food items available for customers in the restaurant.

6. Order:

- **Attributes:** order_id, customer_id (FK), table_id (FK), food_item_id (FK), order_date, total_price, status
- **Description:** Manages orders placed by customers in the restaurant.

7. Inventory:

- **Attributes:** inventory_id, item_name, quantity, unit_price, last_restocked_date
- **Description:** Tracks the stock of food items and room supplies.

8. Menu:

- **Attributes:** menu_id, food_item_id (FK), date_added, special_offers
- **Description:** Manages the food items listed on the menu for different meals (breakfast, lunch, dinner).

9. Bill:

- **Attributes:** bill_id, reservation_id (FK), order_id (FK), total_amount, bill_date, payment_status
- **Description:** Tracks the bills generated for both room reservations and dining orders.

10. Payment:

- **Attributes:** payment_id, bill_id (FK), payment_method, payment_date, amount
- **Description:** Tracks payments made by customers for services, rooms, or dining orders.

11. Table:

- **Attributes:** table_id, seating_capacity, status, location
- **Description:** Manages information about tables in the restaurant (available, reserved, or occupied).

12. Feedback:

- **Attributes:** feedback_id, customer_id (FK), feedback_text, rating, feedback_date
- **Description:** Stores customer feedback on services, rooms, or food quality.

13. Restaurant:

- **Attributes:** restaurant_id, name, location, opening_hours
- **Description:** Tracks details about the hotel restaurant, including location and hours of operation.

14. ServiceRequest:

- **Attributes:** request_id, customer_id (FK), service_type, status, requested_date
- **Description:** Manages requests made by customers (e.g., room cleaning, food delivery).

15. Event:

- **Attributes:** event_id, customer_id (FK), event_name, event_date, location, event_price
- **Description:** Manages events hosted by the hotel such as conferences, weddings, and other gatherings.

Modules and Functionalities:

1. Room Management:

- **Description:**
 - Manage room availability, reservations, check-ins, and check-outs.
 - Assign rooms based on customer preferences.
 - Track room status (available, occupied, under maintenance).
- **Key Features:** Room booking, cancellation, updates to room status, and room services.

2. Restaurant Management:

- **Description:**
 - Manage food orders, menu updates, table reservations, and dining preferences.
 - Process customer orders for dine-in or room service.
 - Handle table availability, booking, and dining schedules.

- **Key Features:** Menu management, order tracking, table reservation, and food item availability.

3. Inventory Management:

- **Description:**
 - Track inventory for food items in the restaurant and room supplies (towels, toiletries).
 - Automatically notify staff when stock levels are low.
- **Key Features:** Inventory updates, restocking, low-stock alerts, and item categorization.

4. Billing and Payment:

- **Description:**
 - Generate bills for hotel stays and dining orders.
 - Manage different payment methods (credit/debit cards, cash, online payments).
 - Handle refunds for cancelled reservations or returned orders.
- **Key Features:** Bill generation, invoice management, and payment processing.

5. Event Management:

- **Description:**
 - Manage bookings and organization for hotel events (e.g., weddings, conferences).
 - Assign event locations within the hotel, such as banquet halls or outdoor spaces.
 - Track event payments and additional services (catering, decorations).
- **Key Features:** Event scheduling, service request management, and event billing.

6. Customer Feedback:

- **Description:**
 - Collect feedback from customers on their stay or dining experience.
 - Analyze feedback to improve services, food quality, and overall experience.
- **Key Features:** Feedback collection, rating system, and reporting for service improvement.

7. Reporting Module:

- **Description:**
 - Generate reports on room occupancy rates, revenue, inventory usage, and food consumption trends.
 - Provide insights into customer satisfaction through feedback analysis.
- **Key Features:** Sales and revenue tracking, inventory reports, and customer insights.

High-Level Workflow:

1. Customer Reservation:

- A customer books a room online or via the hotel desk.
- The room availability is checked, and the reservation is confirmed.
- The customer checks in and can request room service or additional services during their stay.

2. Dining and Table Reservation:

- The customer makes a dining reservation at the restaurant or orders room service.
- The staff processes the order, and the customer is served.
- Upon completion of the meal, the customer is billed, and payment is processed.

3. Event Booking:

- A customer books a banquet hall or outdoor space for a wedding or conference.
- The event is organized, and necessary services are provided.
- Billing is generated post-event, and payment is completed.

4. Inventory Replenishment:

- The system monitors inventory levels of food items and room supplies.
- When stock is low, restocking is initiated.
- Inventory is updated once new stock arrives.

5. Billing and Payment:

- Bills are generated for all services (rooms, dining, events).
- Customers make payments via various methods.
- Payment status is tracked and updated in the system.

This requirement analysis provides a complete blueprint for developing a **Hotel and Restaurant Management System** using Core Java, Collections, and Hibernate.

Detailed Modules Description

The Hotel and Restaurant Management System includes several interconnected modules, each responsible for specific operations related to the hotel and restaurant business. Below is a detailed description of each module, outlining its functions and features:

1. Room Management Module:

Purpose:

To handle all operations related to hotel room management, including reservations, availability, and services provided to guests during their stay.

Key Features:

- **Room Availability Tracking:** Displays available, occupied, or under-maintenance rooms in real-time.
 - **Reservation Management:** Facilitates the booking of rooms by guests either online or in person. Handles room allocation based on customer preferences, including dates, room type, and any special requests.
 - **Check-In/Check-Out Management:** Handles guest check-ins and check-outs, updating room status accordingly.
 - **Room Services:** Manages additional services like cleaning, laundry, or food delivery requested by the guest during their stay.
 - **Room Status Updates:** Tracks real-time updates of room conditions such as occupancy or maintenance requirements.
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2. Restaurant Management Module:

Purpose:

To manage the operations of the hotel restaurant, including dining orders, table bookings, and menu management.

Key Features:

- **Menu Management:** Allows the restaurant to create and update the list of available food items for different times of the day (breakfast, lunch, dinner). Includes managing special offers or discounts on food items.
 - **Order Management:** Handles food orders from the hotel restaurant or room service, including order placement, tracking, and preparation status.
 - **Table Reservation:** Manages table bookings, tracking table availability, and assigning tables based on customer preferences (e.g., party size or location within the restaurant).
 - **Order Status Updates:** Tracks the status of orders, from preparation to serving, and updates both staff and customers on the progress.
 - **Bill Generation:** Generates bills for customers based on the food items ordered and any applicable taxes or discounts.
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3. Inventory Management Module:

Purpose:

To ensure smooth stock management of food and supplies, both for restaurant operations and hotel room services.

Key Features:

- **Stock Level Tracking:** Continuously monitors the stock of food items in the restaurant and supplies (towels, toiletries, etc.) required for room services.
 - **Low-Stock Alerts:** Notifies the staff when stock levels fall below a predefined threshold to avoid shortages.
 - **Restocking Management:** Facilitates the process of ordering and replenishing stock when necessary.
 - **Inventory History:** Maintains a record of inventory usage, helping to analyze consumption trends and plan for future requirements.
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4. Billing and Payment Module:**Purpose:**

To manage billing for hotel reservations, room services, and dining orders, along with payment processing.

Key Features:

- **Bill Generation:** Generates detailed bills for customers, including room charges, food orders, services (e.g., laundry, event charges), taxes, and applicable discounts.
 - **Payment Processing:** Supports various payment methods such as credit/debit cards, online payment gateways, and cash. Integrates with secure payment systems.
 - **Refund Management:** Handles refunds for cancelled reservations or returned food orders.
 - **Invoice Management:** Stores a record of all invoices generated, enabling staff to retrieve and manage them for accounting purposes.
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5. Event Management Module:**Purpose:**

To organize and manage hotel events, including conferences, weddings, or parties.

Key Features:

- **Event Booking:** Allows customers to book event spaces (banquet halls, conference rooms, outdoor areas) based on the type and size of the event.
 - **Service Coordination:** Manages event services such as catering, decorations, and staff required for the event.
 - **Event Scheduling:** Coordinates event timings and resource allocation to avoid conflicts with other bookings.
 - **Billing and Payment:** Generates a bill for event services based on the space, catering, and other requested services.
 - **Event Tracking:** Keeps track of event details such as customer preferences, guest count, and services provided.
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6. Customer Feedback Module:

Purpose:

To collect, manage, and analyze customer feedback to improve the quality of services.

Key Features:

- **Feedback Collection:** Allows customers to submit feedback on their hotel stay or restaurant dining experience through online forms or on-site surveys.
 - **Rating System:** Enables customers to rate various aspects such as room cleanliness, service quality, and food taste.
 - **Feedback Analysis:** Analyzes feedback to identify trends and areas of improvement for the hotel and restaurant.
 - **Staff Evaluation:** Allows management to assess staff performance based on customer feedback.
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7. Service Request Module:**Purpose:**

To handle customer requests for room services such as cleaning, laundry, or food delivery.

Key Features:

- **Request Submission:** Enables customers to submit service requests either via an online portal or through direct communication with hotel staff.
 - **Request Status Tracking:** Provides real-time updates to customers on the status of their requests (e.g., cleaning in progress, food delivered).
 - **Service History:** Maintains a log of all service requests made by a customer during their stay, helping staff anticipate future needs.
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8. Reporting Module:**Purpose:**

To generate reports that provide insights into hotel and restaurant performance, helping management make informed decisions.

Key Features:

- **Revenue Reports:** Generates reports on total revenue generated from room bookings, dining services, and events.
 - **Occupancy Reports:** Tracks room occupancy rates and trends over time, helping to optimize pricing and promotions.
 - **Food Consumption Reports:** Analyzes consumption patterns in the restaurant, enabling better inventory planning and menu updates.
 - **Customer Feedback Reports:** Summarizes customer feedback to identify key strengths and areas that need improvement.
 - **Staff Performance Reports:** Provides an evaluation of staff performance based on feedback and request handling.
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9. User Management Module:

Purpose:

To manage user roles and access permissions, ensuring that different users have appropriate access to various parts of the system.

Key Features:

- **Role-Based Access:** Ensures that different users (administrators, staff, customers) have access to only the functionalities they need.
- **User Account Management:** Allows administrators to create, update, or delete user accounts and reset passwords.
- **User Activity Logs:** Tracks user activities, helping to monitor system usage and detect potential security issues.

These modules work together to ensure seamless operations in the hotel and restaurant, from managing rooms and inventory to handling events, payments, and customer feedback.

Detailed Workflow

The workflow in the Hotel and Restaurant Management System is composed of a series of interconnected processes that manage hotel room reservations, restaurant orders, inventory management, billing, and customer feedback. Each module contributes to a smooth and efficient operation by interacting with other modules. Below is the detailed step-by-step workflow for key operations within the system:

1. Room Reservation Workflow

Objective: To allow customers to reserve rooms online or in person.

1. Customer Browses Available Rooms:

- The customer views available rooms on the hotel's online booking platform or through the receptionist.
- The system retrieves room availability from the **Room Management Module** based on check-in and check-out dates, room type, and occupancy.

2. Customer Selects Room:

- The customer selects a room based on availability and preferences (e.g., single, double, suite).

3. Reservation Details Entered:

- The customer enters personal details (name, contact information, etc.), stay duration, and any special requests.
- The system calculates the total room cost based on the stay duration and room rate.

4. Reservation Confirmation:

- The reservation is confirmed by making a partial/full payment via the **Payment Gateway Module**.
- The **Reservation Entity** stores all relevant details in the database, including customer details, room allocation, and payment status.

5. Room Allocation and Check-In:

- On the check-in date, the system retrieves the reservation details and marks the room as occupied.
 - The **Room Management Module** updates the room's status to "occupied," and the **Bill Module** starts tracking charges related to the stay.
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2. Restaurant Order Workflow

Objective: To manage the process of placing and fulfilling dining orders within the hotel restaurant or via room service.

1. Customer Places Order:

- The customer browses the available food items through a menu provided by the **Menu Management Module**.
- Customers can either place orders for dine-in (by reserving a table) or request room service.

2. Order Processing:

- The system validates the order, checking for item availability from the **Inventory Management Module**.
- If items are available, the order is placed, and the kitchen staff is notified via the **Order Management Module**.

3. Order Preparation:

- The kitchen staff prepares the order. The order status is updated in the system as "In Progress."
- Once the order is ready, the status is updated to "Ready for Service."

4. Order Serving:

- For dine-in customers, the **Table Entity** is updated with the order details.
- For room service, the room number is assigned, and the **ServiceRequest Module** coordinates delivery.

5. Bill Generation and Payment:

- After the meal is completed, the bill is generated using the **Billing and Payment Module**. The bill includes details like food items ordered, taxes, discounts, and service charges.
- Payment is processed via the **Payment Gateway Module** using various payment options (cash, credit card, digital payment, etc.).

3. Inventory Management Workflow

Objective: To ensure that stock levels for food items and room supplies are maintained.

1. Stock Level Monitoring:

- The system tracks the availability of food items, beverages, and hotel supplies (towels, toiletries, etc.) in real-time via the **Inventory Management Module**.

2. Automatic Low-Stock Alerts:

- When stock levels fall below a predefined threshold, the system automatically generates low-stock alerts, notifying the inventory manager to replenish stock.

3. Restocking and Purchase Orders:

- The inventory manager creates a purchase order for required items and updates the inventory once the stock is received.
- The **Inventory Entity** is updated with new stock levels.

4. Inventory Consumption:

- The system tracks item usage based on room service requests, restaurant orders, and general hotel operations.
- The consumption data is fed into the **Reporting Module** to generate reports on trends and projections.

4. Billing and Payment Workflow

Objective: To handle the generation of bills for various services (rooms, dining, events) and process payments.

1. Bill Generation:

- The system calculates the total charges for room bookings, food orders, and additional services like laundry or event bookings.
- The **Billing Module** consolidates all charges (room rate, service charges, taxes, etc.) and generates an invoice.

2. Payment Processing:

- Customers can review their bill and proceed to payment through the **Payment Gateway Module**, which supports multiple payment methods.
- Once payment is made, the **Payment Entity** is updated, and a payment receipt is generated.

3. Refunds (if applicable):

- In case of cancellations or service issues, the **Refund Workflow** is triggered to reimburse the customer according to the hotel's refund policy.

5. Event Management Workflow

Objective: To handle bookings and management of hotel events (conferences, weddings, etc.).

1. Event Inquiry:

- The customer browses available event spaces and services (catering, decorations) via the **Event Management Module**.
- The system retrieves event space availability and provides a quotation based on the customer's preferences (number of guests, type of event, etc.).

2. Event Booking:

- The customer confirms the booking, providing event details like date, time, and additional services needed.
- A deposit payment is made via the **Payment Gateway Module**, and the event is added to the system's schedule.

3. Service Coordination:

- The system assigns staff and resources (caterers, decorators) to the event.
- Coordination between the **ServiceRequest Module** and event organizers ensures smooth event execution.

4. Post-Event Billing:

- After the event, the final bill is generated, which includes any additional charges for extended hours or extra services.
- Payment is processed, and an invoice is issued.

6. Feedback Workflow

Objective: To collect feedback on the customer's hotel stay and dining experience.

1. Feedback Request:

- After check-out or upon completion of a meal, the system sends a feedback request to the customer via email or an in-app form.

2. Customer Submits Feedback:

- The customer rates various aspects like room service, cleanliness, food quality, and staff behavior using the **Feedback Module**.

3. Feedback Analysis:

- The system stores and analyzes the feedback data, generating insights for the management team.
- Negative feedback triggers a review process, where management can take corrective action and respond to the customer.

4. Reporting:

- The **Reporting Module** summarizes feedback and provides performance evaluations of staff and services based on customer input.

7. Reporting Workflow

Objective: To generate reports for hotel management, providing insights into operations and customer experiences.

1. Data Collection:

- The system collects data from various modules (billing, reservations, inventory, feedback, etc.).

2. Report Generation:

- The **Reporting Module** generates different types of reports, such as:
 - **Revenue Reports:** Summarizes income from rooms, events, and restaurant services.
 - **Occupancy Reports:** Shows hotel room occupancy trends over time.
 - **Food Consumption Reports:** Details food usage in the restaurant, helping with inventory planning.
 - **Feedback Reports:** Provides insights into customer satisfaction levels.

3. Report Review:

- Hotel management reviews these reports to make data-driven decisions, improve service quality, and optimize resources.

8. Staff and User Management Workflow

Objective: To manage user roles, access control, and staff performance.

1. User Role Assignment:

- Administrators assign roles to staff members, determining their access levels within the system (e.g., reception staff, kitchen staff, management).

2. Access Control:

- The system ensures that users can only access the functionalities relevant to their roles, providing security and control over operations.

3. Activity Tracking:

- The system logs user actions to monitor staff performance and ensure accountability.
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These workflows ensure efficient operations in the hotel and restaurant, optimizing customer service and internal processes across multiple modules.